National Energy and Emission Modeling and Analysis Tool

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Input Data | Analysis | Results

Start

The tool description will be added here.
The content will be provided later.

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Input Data | Analysis | Results

- Vehicle Classification Data
- Projected Vehicle Penetration Rate Data
- Traffic Volume and Speed
- Projected Demand

Upload Vehicle Classification Data

Input the Base Year

202#

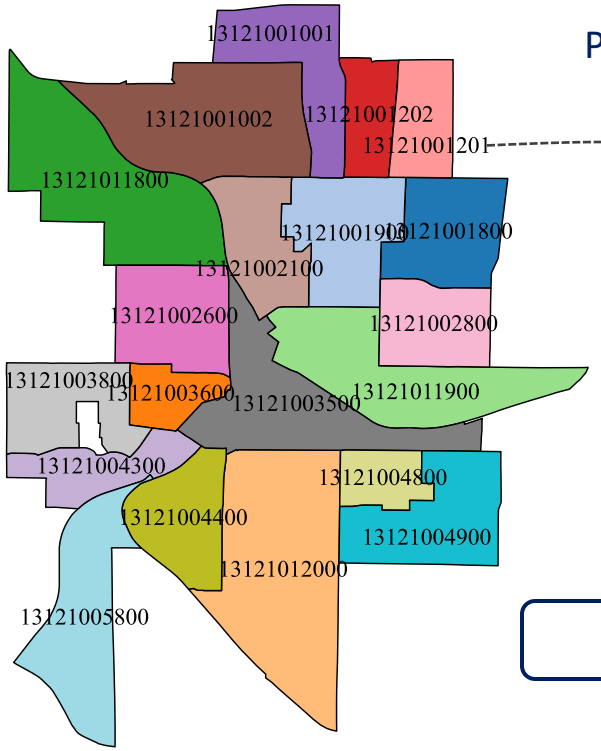
Vehicle Type

City

Age	Penetration Rate					
	Gasoline	E-85	Diesel	CNG	Electricity	Total
0						
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						
17						
18						
19						
20						
21						
22						
23						
24						

From the uploaded data file (modifiable by the user)

Check: the sum of all percentages across vehicle type, fuel type, and year must be exactly 100%.



Only the last 4 digits of the tract number on the map

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- Vehicle Classification Data
- Projected Vehicle Penetration Rate Data
- Traffic Volume and Speed
- Projected Demand

From 0 to 5. Year 0 is the "Base Year" provided in the previous page

Upload Projected Vehicle Penetration Rate Data

Projection Year | Vehicle Type

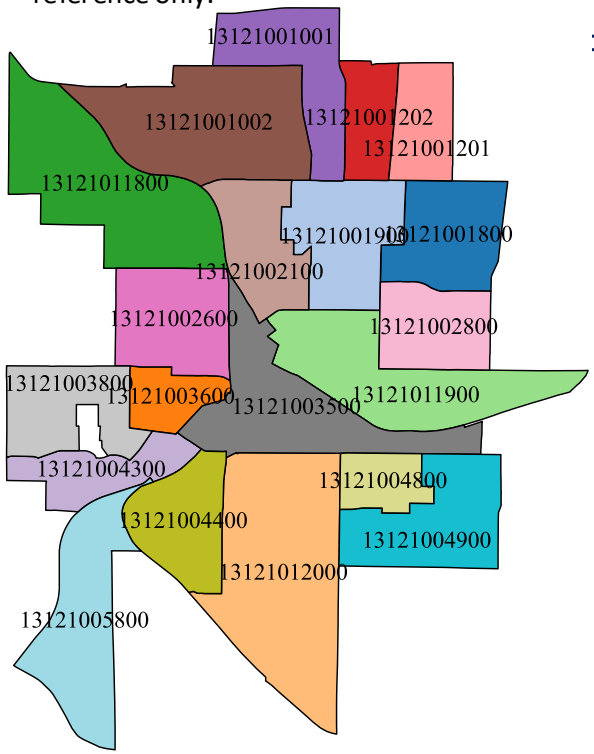
City Name

The city cannot be changed at this step; it is displayed for user's reference only.

Year	Penetration Rate					
	Gasoline	E-85	Diesel	CNG	Electricity	Total
0						
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						
17						
18						
19						
20						
21						
22						
23						
24						

From the uploaded data file (modifiable by the user)

Check: the sum of all percentages across vehicle type, fuel type, and year must be exactly 100%.



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Input Data | Analysis | Results

- Vehicle Classification Data
- Projected Vehicle Penetration Rate Data
- Traffic Volume and Speed
- Projected Demand

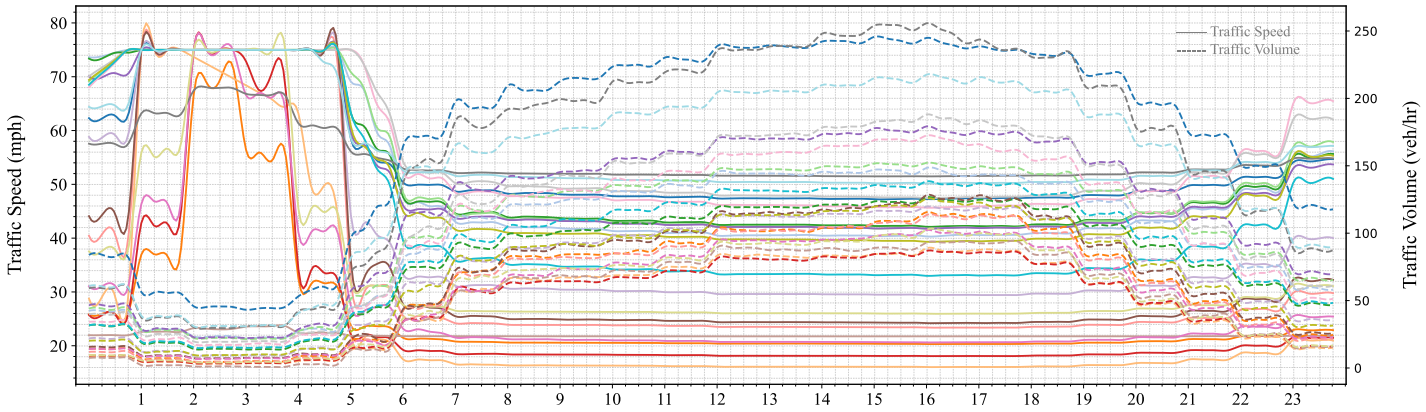
Upload Traffic Volume Data

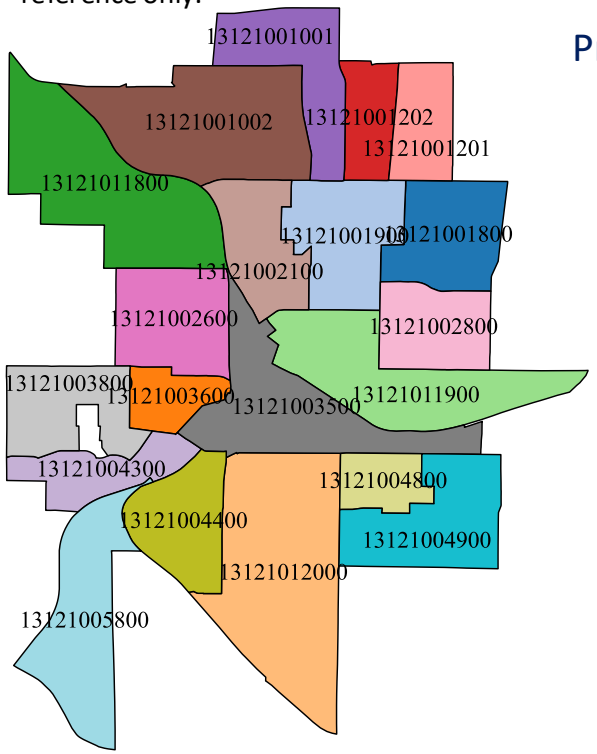
Volumes for Year 0 (base year)

Upload MFD Parameters

City Name

The city cannot be changed at this step; it is displayed for user's reference only.





Census Tract	Model Parameters				
	V_f	Q_m	k_j	λ	w
Tract #	From the uploaded data file (modifiable by the user)				

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Input Data | Analysis | Results

- Vehicle Classification Data
- Projected Vehicle Penetration Rate Data
- Traffic Volume and Speed
- Projected Demand

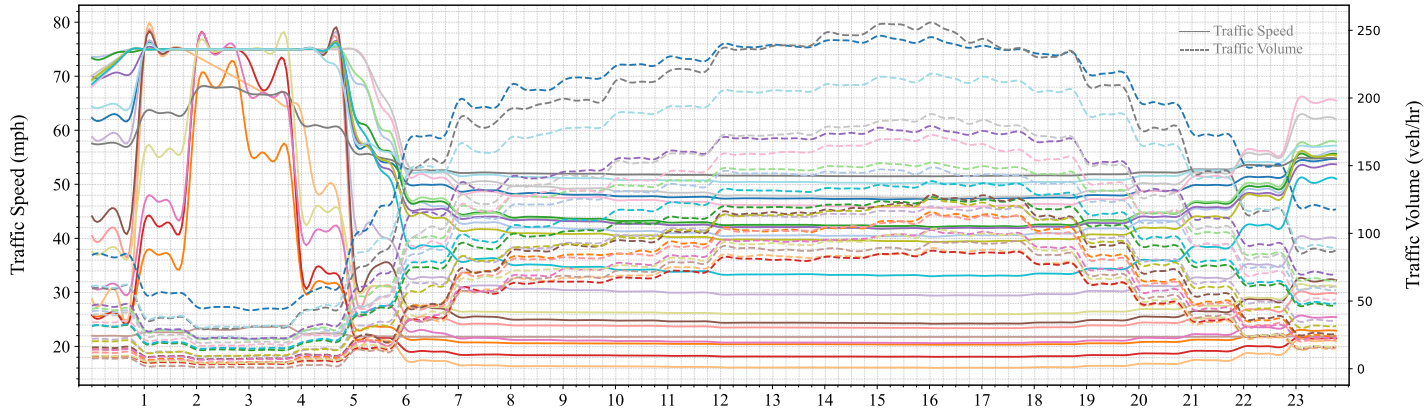
Upload Projected Traffic Volume Data

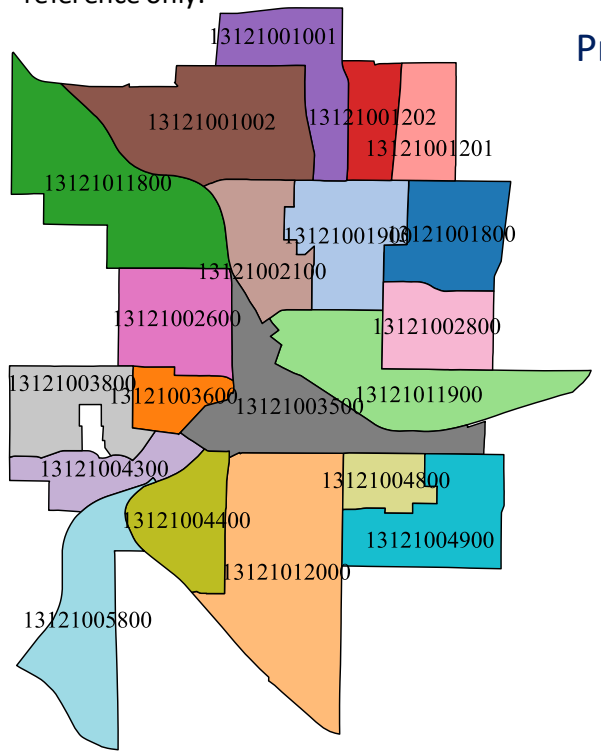
From 0 to 5. Year 0 is the "Base Year" provided in the previous page

Projection Year

City Name

The city cannot be changed at this step; it is displayed for user's reference only.





Years will be based on the user's input for the base year on the first page

Census Tract	Projected Increase in Traffic Volumes				
	202#+1	202#+2	202#+3	202#+4	202#+5
Tract #	From the uploaded data file (modifiable by the user)				

Census Tract	Model Parameters				
	V_f	Q_m	k_j	λ	w
Tract #	The parameters cannot be changed at this step; it is displayed for user's				

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Input Data | Analysis | Results

Vehicle Energy Consumption and Emission Rates

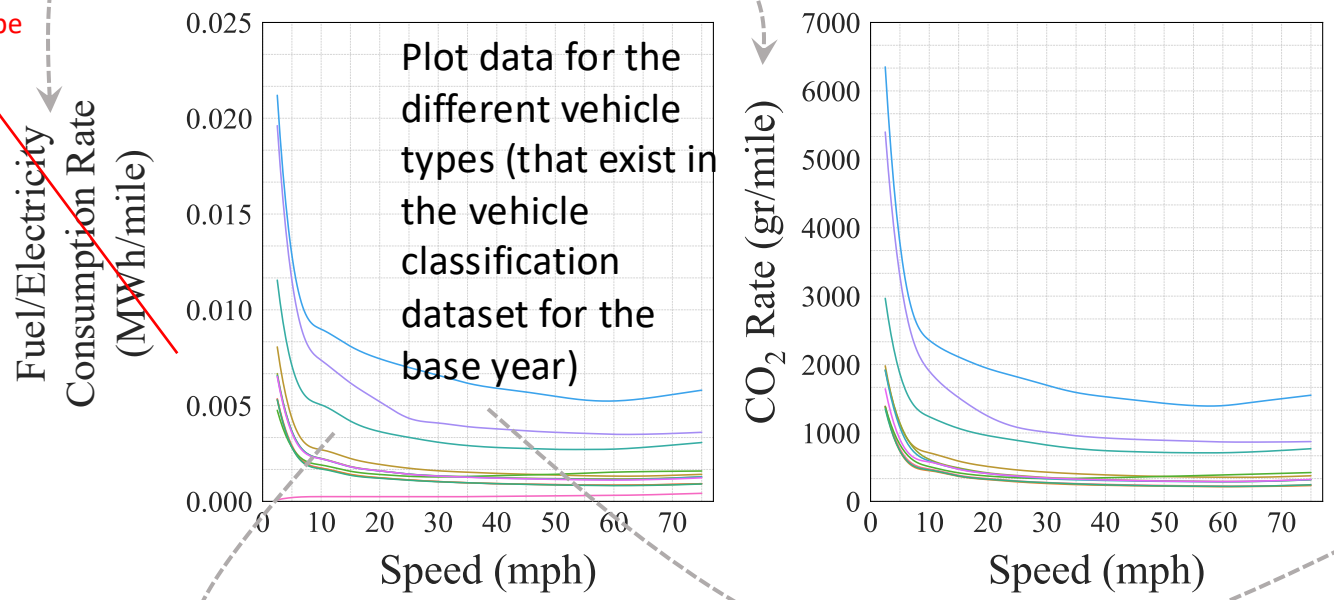
Grid Emission Rates

The unit of fuel consumption in the model is MWh but it can be simply converted to Gallon using a conversion factor for each fuel type

Fuel Type | Emission Type | Vehicle Age

City Name

The city cannot be changed at this step; it is displayed for user's reference only.



Nueral Network Model, NOT the raw data!

- Passenger Car -
- Passenger Car -
- Light Commerc
- Passenger Trucl
- Motorcycle -
- Passenger Ca
- Single Unit S
- Passenger Tri
- Combination Lc
- Single Unit Sho
- Light Commerc
- Passenger Car -

Powered by: MOVES Motor Vehicle Emission Simulator

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Scenarios Projecting Technology Costs, Fuel Prices, Demand Growth, and Electricity Sector Policies

Mid-case: This scenario represents central estimates for key inputs such as technology costs, fuel prices, and demand growth, with no inclusion of nascent technologies. It assumes electric sector policies as they existed in September 2023.

Low Renewable Energy and Battery Costs: This scenario mirrors the assumptions of the Mid-case but assumes lower costs and greater performance improvements for renewable energy and batteries.

High Renewable Energy and Battery Costs: Similar to the Mid-case scenario, but with higher costs and less significant performance improvements for renewable energy and batteries.

High Electricity Demand Growth: This scenario also uses the same base assumptions as the Mid-case but assumes a higher average annual demand growth rate of 2.8% from 2024 to 2050, consistent with a trajectory towards 100% economy-wide decarbonization.

Low Natural Gas Prices: This scenario is consistent with the Mid-case assumptions but assumes lower natural gas prices.

High Natural Gas Prices: This scenario follows the same base assumptions as the Mid-case but assumes higher natural gas prices.

95% Decarbonization by 2050: Building on the Mid-case scenario, this assumes the inclusion of nascent technologies and enforces a national decarbonization constraint, progressively reducing net electricity sector emissions to 5% of 2005 levels by 2050.

100% Decarbonization by 2035: Similar to the 95% Decarbonization scenario, but with a more aggressive target of achieving net-zero emissions by 2035 through the inclusion of nascent technologies and a linear reduction to zero emissions.

Vehicle Energy Consumption and Emission Rates

City Name

Grid Emission Rates

Powered by: NREL Cambium

Grid Emission Type

CO2 Emission (kg/MWh)

ALL 8 scenarios (use Faysal's template for plotting)

202# 202#+1 203#+5

Mid-case

Legend here

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These graphs should be generated for the specified year range based on the user's input reference year on page one, using the Prophet model, NOT the raw Cambium data.

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Notifications

Input Data | Analysis | Results

Generate Results

The unit of fuel consumption in the model is MWh but it can be simply converted to Gallon using a conversion factor for each fuel type

Gallon for fuel/MWh for electricity)

Fuel Type

ALL city tracts color coded with the map (use Faysal's speed volume profile template for plotting)

1:00 23:00 24:00

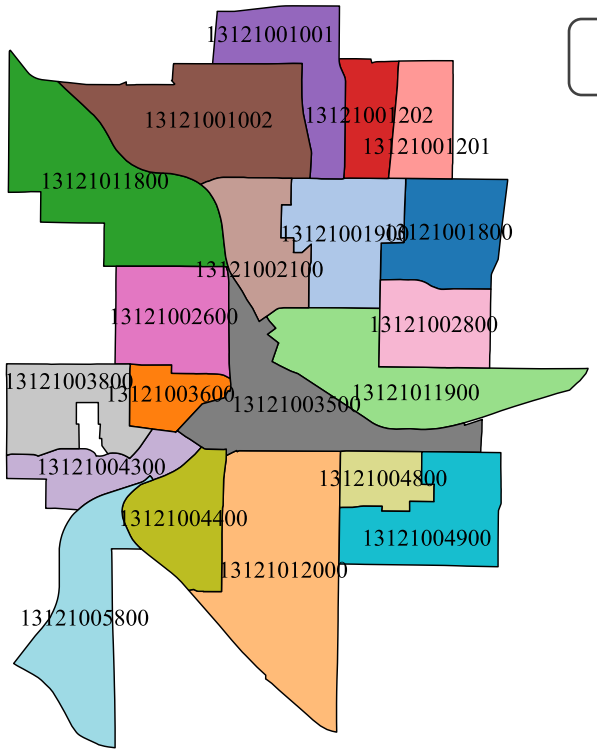
CO2 (ton)

Emission Type

ALL city tracts color coded with the map (use Faysal's speed volume profile template for plotting)

1:00 23:00 24:00

City Name



Vehicles / Grid

Daily / Annual

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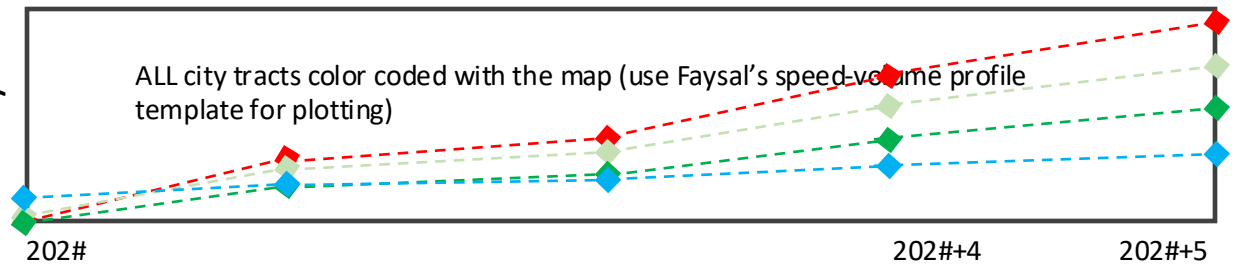
Notifications

Input Data | Analysis | Results

The unit of fuel consumption in the model is MWh but it can be simply converted to Gallon using a conversion factor for each fuel type

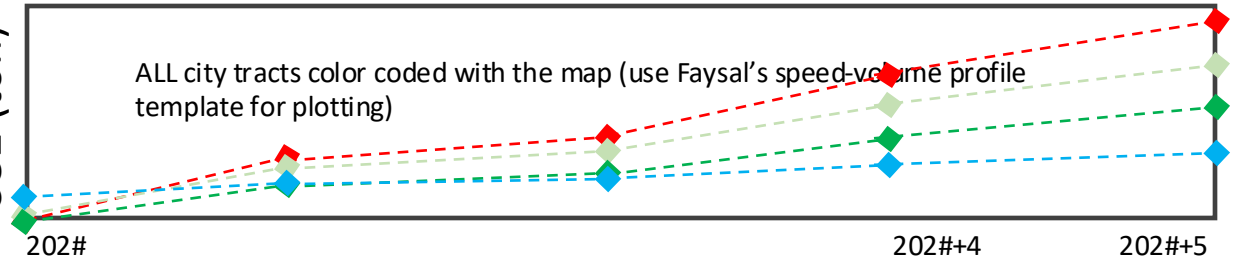
Gallon for fuel/MWh for electricity)

Fuel Type

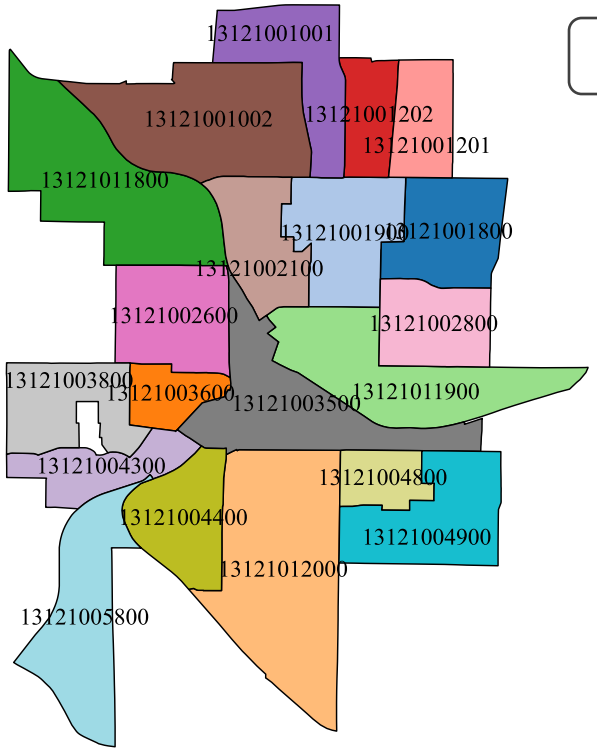


CO2 (ton)

Emission Type



City Name



Vehicles / Grid

Daily / Annual

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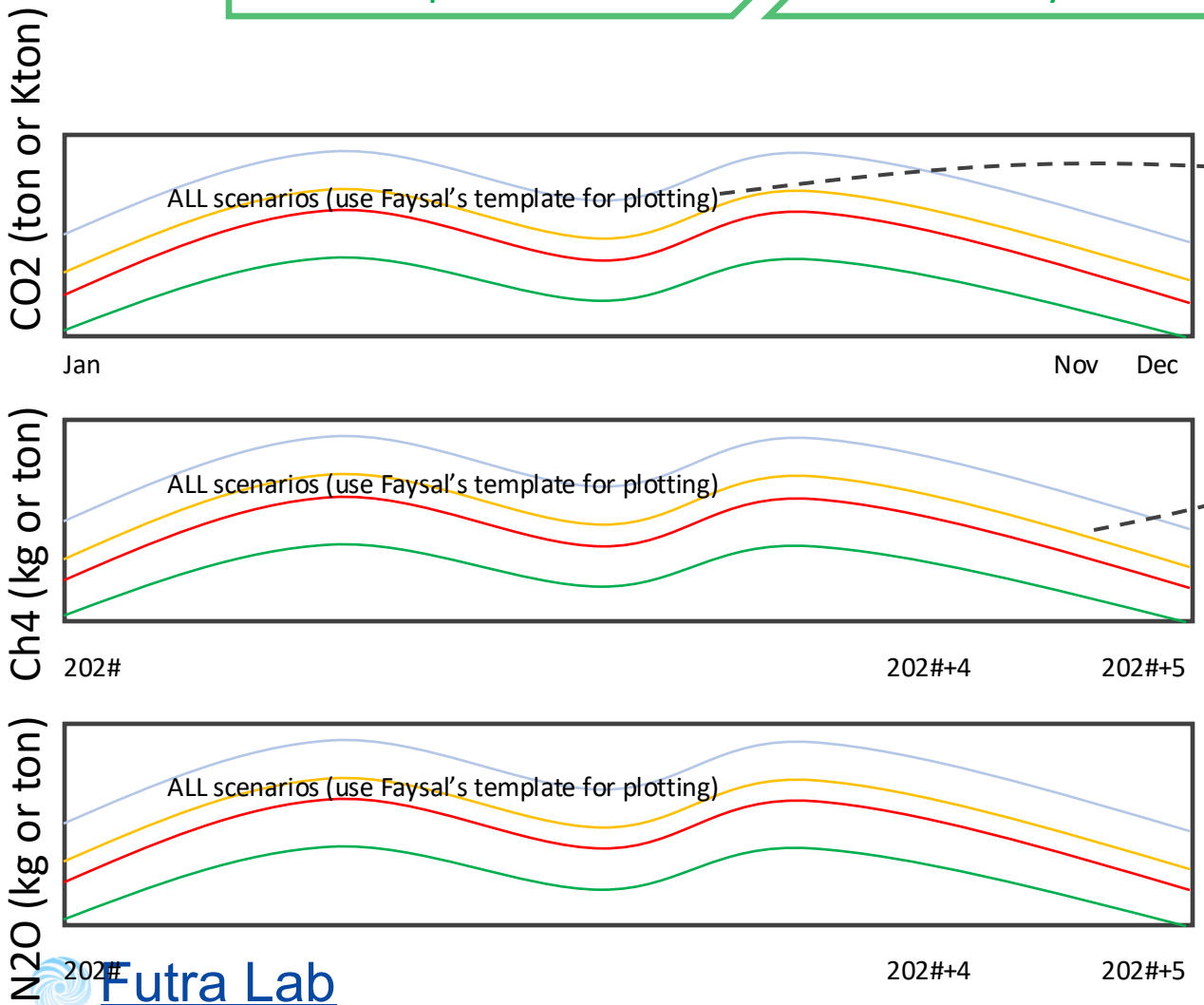
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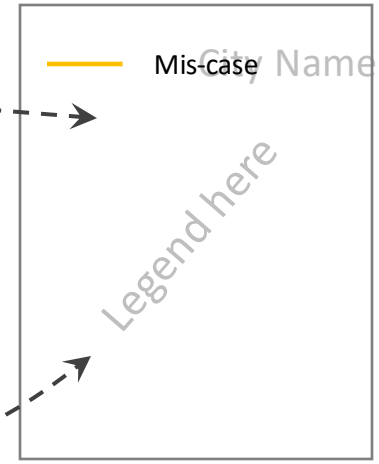
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Vehicles / Grid

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